

Auditing the Judge: Internal Signals for LLM Reasoning Assurance Supplementary Material

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APPENDIX A PROOFS AND NUMERICAL VERIFICATION

The appendix first proves five results and then checks the three finite-sample identities against the recorded artifacts. Their scopes are:

Result	Type of claim	Where used
Residual risk (Proposition 1)	Exact finite-sample identity	Section III-C Section VI-B
AUROC (Proposition 2)	Population identity with exact finite-sample analogue	Section III-D Section VI
Fisher direction (Proposition 3)	Model-dependent population result (shared-covariance Gaussian)	Section IV-C
Residual shift (Lemma 1)	Exact algebraic decomposition	Section IV-E Section VII
Routing identities	Exact finite-sample identities	Section III-E Section VI-C

Here, “exact” means that the result follows from the cited definitions without a distributional model. Keeping proofs and artifact checks separate prevents numerical agreement from being mistaken for proof.

A. Proofs

1) *Residual-risk identity (Proposition 1)*: The proof expresses both metrics through the same quantity: the expected number of errors selected for review.

Proof. Work with the finite-sample quantities on a population of N items containing $W = \pi_0 N$ errors, and let the budget select M_m items, with realized fraction $b_m = M_m/N$. For a random resolution of any cutoff tie, let C be the number of selected errors and let $\bar{C} := \mathbb{E}_{\text{tie}}[C]$. Every tie resolution selects exactly M_m items. Hence, by Equations (4) and (5),

$$\mathcal{R}_m = \frac{\bar{C}}{W}, \quad \varrho_m = \frac{W - \bar{C}}{N - M_m}.$$

Substituting $\bar{C} = W\mathcal{R}_m$, $W = \pi_0 N$, and $M_m = b_m N$ gives

$$\varrho_m = \frac{W(1 - \mathcal{R}_m)}{N - M_m} = \frac{\pi_0 N(1 - \mathcal{R}_m)}{N(1 - b_m)} = \pi_0 \frac{1 - \mathcal{R}_m}{1 - b_m}.$$

For two scores on the same population and at the same realized budget, π_0 and b_m are common. Subtracting the identity for s from the identity for s' therefore gives Equation (7), including its sign. $\square$

Thus recall and residual risk are not independent measurements: for fixed π_0 and b_m , either one determines the other exactly. The artifact check is reported in Section A-B1.

2) *AUROC identity (Proposition 2)*: The proof decomposes each increase in flagged population mass into error mass and correct-answer mass, then integrates error recall along that path.

Proof. Let $D = -s$ and treat errors as the positive class, so flagging the lowest-scoring items is equivalent to thresholding D from above. Along the randomized threshold path, write TPR and FPR for the error-class and correct-class selection rates, respectively. Uniform randomization within a tied score group fills in the corresponding line segment of the ROC curve. At every point on this path, the flagged population fraction is, by the law of total probability,

$$m = \pi_{0,P} \text{TPR} + (1 - \pi_{0,P}) \text{FPR}. \quad (21)$$

As the path moves from selecting no item to selecting every item, m , TPR, and FPR each move monotonically from 0 to 1. Thus Equation (21) parameterizes the randomized ROC curve by budget, and, in the Riemann–Stieltjes sense, $dm = \pi_{0,P} d\text{TPR} + (1 - \pi_{0,P}) d\text{FPR}$. Since $\mathcal{R}_m^P = \text{TPR}$ along this path,

$$\begin{aligned} \int_0^1 \mathcal{R}_m^P dm &= \pi_{0,P} \int_0^1 \text{TPR} d\text{TPR} \\ &\quad + (1 - \pi_{0,P}) \int_0^1 \text{TPR} d\text{FPR} \\ &= \frac{\pi_{0,P}}{2} + (1 - \pi_{0,P}) \text{AUROC}_P. \end{aligned}$$

The first integral is $\frac{1}{2}$ because TPR runs from 0 to 1. The second is the ROC area of D when errors are positive. With

half credit for ties, this equals the ROC area of s when correctness is positive:

$$\begin{aligned} \mathbb{P}(D_0 > D_1) + \frac{1}{2}\mathbb{P}(D_0 = D_1) \\ = \mathbb{P}(s_1 > s_0) + \frac{1}{2}\mathbb{P}(s_1 = s_0), \end{aligned}$$

where s_0 and s_1 are the scores of independent error-class and correct-class draws, respectively, and $D_c = -s_c$. This is $\text{AUROC}_P(s)$ and establishes Equation (8). $\square$

The finite-sample correction is also exact. Assume $0 < W < N$, randomly order items within each tied score group, and let r_e be the resulting rank of error item e in ascending score order. The item contributes to the captured-error count at every budget j/N for which $j \geq r_e$. Therefore

$$\frac{1}{N} \sum_{j=1}^N \mathcal{R}_{j/N} = \frac{1}{NW} \sum_{e: y_e=0} \mathbb{E}_{\text{tie}}[N - r_e + 1].$$

For every error–correct pair, the probability that the correct item follows the error item is precisely its AUROC contribution, including half credit for a tie. These pairs therefore contribute $W(N - W)\text{AUROC}$ in expectation. Every unordered pair of errors contributes one later error, giving $W(W - 1)/2$, and each error also contributes itself, giving W . Dividing the total by NW yields

$$\frac{1}{N} \sum_{j=1}^N \mathcal{R}_{j/N} = \frac{\pi_0}{2} + (1 - \pi_0)\text{AUROC} + \frac{1}{2N}. \quad (22)$$

The $1/(2N)$ term is the finite-grid endpoint correction; it vanishes as $N \rightarrow \infty$ and is absent from the population integral in Equation (8).

The corresponding artifact check is reported in Section A-B2.

3) *Population Fisher direction (Proposition 3)*: This proposition explains when the Fisher criterion ranks heads by a population separation parameter. It does not assert that the Gaussian model holds for the observed activations.

Proposition 3 (Population Fisher direction under a shared Gaussian covariance). *For $c \in \{0, 1\}$, let $A_{:,j} \mid y = c \sim \mathcal{N}(\mu_j^c, C_j)$, where C_j is a positive-definite population covariance shared by the two classes. Let $d_j = \mu_j^1 - \mu_j^0$ and the unregularised population Fisher direction $v_j = C_j^{-1}d_j$. If Φ is the standard normal cumulative distribution function and $\delta_j = (d_j^\top C_j^{-1} d_j)^{1/2}$, then*

$$\text{AUROC}(y, A_{:,j}v_j) = \Phi\left(\frac{\delta_j}{\sqrt{2}}\right). \quad (23)$$

Proof. Let $U_c = A_c^\top v_j$, where $A_c \sim \mathcal{N}(\mu_j^c, C_j)$ are independent class-conditional draws. In both classes, the projected variance is $v_j^\top C_j v_j = \delta_j^2$, while the mean difference is $v_j^\top d_j = \delta_j^2$.

First consider the degenerate separation $\delta_j = 0$. Positive definiteness of C_j implies $d_j = 0$ and hence $v_j = 0$. The projected score is constant, so its AUROC is $1/2$ under the half-credit tie convention; this equals $\Phi(0)$.

Now suppose $\delta_j > 0$. Independence gives $U_1 - U_0 \sim \mathcal{N}(\delta_j^2, 2\delta_j^2)$. Because Gaussian draws tie with probability zero, the pairwise definition of AUROC gives

$$\text{AUROC} = \mathbb{P}(U_1 - U_0 > 0) = \Phi\left(\frac{\delta_j^2}{\sqrt{2}\delta_j}\right) = \Phi\left(\frac{\delta_j}{\sqrt{2}}\right).$$

Together, the two cases prove Equation (23). Since Φ is strictly increasing, population AUROC is strictly increasing in the nonnegative Mahalanobis separation δ_j . $\square$

The simplification in Equation (23) does not apply exactly to the implemented regularised $w_j = (\hat{C}_j + \lambda I)^{-1}\hat{d}_j$ with $\lambda = 10^{-2}$; Equation (16) is an empirical train-only selection score.

4) Residual shift (Lemma 1):

Proof. Start from the residual formed on target-domain data using the source-domain map. Adding and subtracting $X_T B_T$ gives

$$R_T = F_T - X_T B_S = \underbrace{(F_T - X_T B_T)}_{R_T^*} + X_T (B_T - B_S).$$

The parenthesized term is exactly the target-map residual R_T^* , and the remaining term is $X_T \Delta B$. This is Equation (20). No distributional assumption is needed for the algebraic identity once the two maps are defined. $\square$

Two qualifications delimit the result. First, it is a decomposition, not a bound on excess risk: the added term can reinforce or cancel other target-domain shift, so the lemma alone predicts neither the sign nor magnitude of a performance change. Second, ΔB is a population object; on finite samples the fitted $\hat{B}$ also carries estimation error. In GSM8K→MATH the primary MLP’s observed raw-minus-controlled target gaps are 0.139–0.241 AUROC, consistent with the mechanism, but the experiment does not provide a paired difference-of-differences interval or isolate residualisation causally.

5) *Routing comparators*: These identities condition on the realized greedy and majority-vote answers. They describe how a routing set aggregates the per-item repair values; they do not assume that a deployable policy can observe those values before routing.

Proof of Equations (10) to (12). Define the correctness indicators

$$\begin{aligned} a_i^{(1)} &:= \mathbf{1}[\hat{a}_i^{(1)} \equiv a^*(x_i)], \\ a_i^{(k)} &:= \mathbf{1}[\hat{a}_i^{(k)} \equiv a^*(x_i)], \end{aligned}$$

so $\gamma_i = a_i^{(k)} - a_i^{(1)}$.

Routing identity. Under routing set $\mathcal{A}$, the released answer is the greedy answer outside $\mathcal{A}$ and the majority-vote answer inside it. Therefore

$$\begin{aligned} \text{Acc}(\mathcal{A}) &= \frac{1}{N} \left(\sum_{i \notin \mathcal{A}} a_i^{(1)} + \sum_{i \in \mathcal{A}} a_i^{(k)} \right) \\ &= \frac{1}{N} \sum_{i=1}^N a_i^{(1)} + \frac{1}{N} \sum_{i \in \mathcal{A}} (a_i^{(k)} - a_i^{(1)}), \end{aligned}$$

which is Equation (10).

Random comparator. If $\mathcal{A}$ is uniform over all subsets of size M_m , then every item has inclusion probability $M_m/N = b_m$. Linearity of expectation gives

$$\mathbb{E}_{\mathcal{A}} \left[\sum_{i \in \mathcal{A}} \gamma_i \right] = \sum_{i=1}^N \gamma_i \mathbb{P}_{\mathcal{A}}(i \in \mathcal{A}) = M_m \bar{\gamma}.$$

Substitution into Equation (10) proves Equation (11). Taking $\mathcal{A} = [N]$ also shows that $\bar{\gamma}$ is the accuracy change produced by routing every item.

Oracle comparator. Because Equation (10) is linear in the selected repair values, a size- M_m maximizer selects them in descending order. The realized population contains $N\pi_+$ values $+1$, $N(1 - \pi_+ - \pi_-)$ zeros, and $N\pi_-$ values -1 . The largest M_m values therefore sum to

$$\min(M_m, N\pi_+) - \max\{0, M_m - N(1 - \pi_-)\}.$$

Dividing by N proves Equation (12). The maximum is first attained at $b_m = \pi_+$, remains constant while zero-valued items are available ($\pi_+ \leq b_m \leq 1 - \pi_-$), and then declines. At $b_m = 1$, its gain over greedy is $\pi_+ - \pi_- = \bar{\gamma}$, as required by route-all accuracy. $\square$

The implemented cost is $1 + k|\mathcal{A}|/N$ solver calls per item: every item receives one initial greedy solve and each routed item receives k new solves. Reusing the greedy answer as one of the k votes would define a different cost and a different experiment.

The corresponding artifact check is reported in Section A-B3.

B. Numerical Verification

These checks ask whether the stored results implement the proved identities. The algebraic decomposition in Lemma 1 requires no data-dependent check. Table IV gives the conclusions before the detailed values.

TABLE IV
SUMMARY OF THE ARTIFACT-LEVEL IDENTITY CHECKS.

Identity	Result
Residual risk	Agreement to floating-point precision in all 42 cells.
AUROC	Difference equals the exact $1/(2N)$ endpoint term to numerical precision.
Routing	Oracle curves agree to floating-point precision in both $k = 5$ arms.

1) *Residual-risk identity:* Equation (7) reproduces every recorded residual-risk improvement from the measured recall gain, prevalence, and realized budget. Agreement is to floating-point precision over all 42 model–dataset–budget cells. Using the nominal budget instead of $b_m = M_m/N$ would create discrepancies up to 1.6×10^{-5} . The six cells at nominal $m = 0.05$ are:

Run	π_0	$\Delta\mathcal{R}$	predicted	recorded
Gemma 3/GSM8K	0.0607	+0.2250	0.0144	0.0144
Qwen2.5/GSM8K	0.0811	+0.0935	0.0080	0.0080
Llama 3.1/GSM8K	0.1783	+0.1073	0.0201	0.0201
Gemma 3/MATH	0.1837	+0.0546	0.0106	0.0106
Qwen2.5/MATH	0.2695	+0.0135	0.0038	0.0038
Llama 3.1/MATH	0.4944	+0.0208	0.0108	0.0108

2) *Finite-sample AUROC identity:* The exact discrete sum agrees with Equation (22) on the aligned held-out predictions, with ties averaged analytically. To display the endpoint term, Table V reports $\text{RHS}_0 := \pi_0/2 + (1 - \pi_0)\text{AUROC}$ without $1/(2N)$; the predicted difference is therefore $\text{LHS} - \text{RHS}_0 = 1/(2N)$.

TABLE V
FINITE-SAMPLE CHECK OF PROPOSITION 2. $\text{LHS} = N^{-1} \sum_j \mathcal{R}_{j/N}$; THE DIFFERENCE FROM RHS_0 SHOULD BE $1/(2N)$.

Run	score	AUROC	LHS	RHS_0
Gemma 3/GSM8K	probe	0.8787	0.8561	0.8557
Gemma 3/GSM8K	judge	0.8484	0.8276	0.8272
Qwen2.5/GSM8K	probe	0.8583	0.8296	0.8293
Qwen2.5/GSM8K	judge	0.8731	0.8432	0.8428
Llama 3.1/GSM8K	probe	0.8127	0.7574	0.7570
Llama 3.1/GSM8K	judge	0.6969	0.6622	0.6618
Gemma 3/MATH	probe	0.9141	0.8381	0.8380
Gemma 3/MATH	judge	0.8221	0.7631	0.7630
Qwen2.5/MATH	probe	0.8878	0.7833	0.7832
Qwen2.5/MATH	judge	0.8479	0.7542	0.7541
Llama 3.1/MATH	probe	0.8647	0.6845	0.6844
Llama 3.1/MATH	judge	0.6955	0.5990	0.5989

The displayed difference is at most 4.0×10^{-4} . Before rounding, it equals $1/(2N) \leq 3.9 \times 10^{-4}$ to numerical precision: the GSM8K rows ($N \approx 1300$) lie near 4×10^{-4} and the MATH rows ($N \approx 4900$) near 1×10^{-4} .

3) *Routing identities:* Both $k = 5$ arms satisfy the oracle identity to floating-point precision. The stored random curve is a Monte Carlo permutation average, so it approximates—rather than exactly equals—the expectation in Equation (11).

On Llama 3.1 8B/GSM8K ($N = 1308$), five-sample voting repairs 120 items and breaks 28. This is the separately generated $k = 5$ voting population, not the $N = 1307$ $k = 1$ discrimination population (Section VIII). Hence $\bar{\gamma} = (120 - 28)/1308$, and the exact random-routing expectation at any realized fraction b_m is $0.8249235 + b_m(92/1308)$. At nominal $m = 0.10$, the implemented cost is approximately 1.50 solver calls per item; the exact expectation and cost use the realized b_m after rounding. Equation (12) predicts a ceiling of $0.8249 + 120/1308 = 0.9167$, first reached at

$\pi_+ = 120/1308 = 0.0917$. The curve remains there while zeros are available, then falls at route-all to 0.8953; the ceiling exceeds route-all by $28/1308 = 0.0214 = \pi_-$.

On Gemma 3 12B/GSM8K ($N = 1319$), voting repairs 19 and breaks 7, on a much higher greedy baseline. The predicted ceiling is $0.9416 + 19/1319 = 0.9560$ at $b_m = 19/1319 = 0.0144$, against a recorded 0.9560 on the plateau; it then falls to route-everything (0.9507) at $b_m = 1$. The ceiling excess is $0.0053 = 7/1319 = \pi_-$. The oracle identity is exact in both arms across an eightfold difference in repairable mass. This agreement checks the artifact implementation; the reason the formula holds is the finite-sample decomposition above, not a fit to these two curves.

The numerical identity checks are reproduced by `scripts/verify_identities.py`.

C. Scope of the Results

The exact identities above explain relationships among the paper’s reported quantities; they are not deployment guarantees. In particular, Lemma 1 does not bound performance under domain shift, and Proposition 3 applies to an unregularised population direction under a shared-covariance Gaussian model, not exactly to the regularised finite-sample selector in Equation (16). Numerical agreement checks the artifact’s implementation of the finite identities but does not replace their proofs.

APPENDIX B

ESTIMATION AND UNCERTAINTY PROTOCOL

This appendix specifies how every reported estimate and uncertainty interval is computed.

A. Tie handling at a budget cutoff

A budget selects an integer M_m of items by score (nearest-integer for the review analysis and floor for routing; Section V-D). When the cutoff falls strictly inside a group of tied scores, the selected set is not determined by the score alone, and resolving the ambiguity by sort order makes a reported number depend on an implementation detail. This can occur for the black-box baseline because finite-precision $p(\text{YES})$ values repeat, although a near-zero or near-one value is not by itself an exact tie (Section C-C).

We report the exact expectation under uniform random tie-breaking. Let the cutoff fall inside a tie group G containing $|G|$ items of which W_G are errors, with j selection slots remaining after all strictly-lower-scoring items are taken. The expected number of errors captured from G is $j W_G / |G|$, so the tie-averaged recall is

$$\mathcal{R}_m = \frac{1}{W} \left(W_{<} + \frac{j W_G}{|G|} \right), \quad (24)$$

with $W_{<}$ the errors among strictly-lower-scoring items. This is the expectation over the randomisation, computed in closed form rather than by sampling, and it is applied identically to every detector. The same convention defines the tie-averaged risk–coverage curves of Section E-B.

B. Bootstrap

Differences between detectors are resampled *paired*: one bootstrap draw selects a multiset of held-out test items, and every detector is re-scored on that same multiset, so the resulting interval reflects the item-level variation the two detectors share rather than treating them as independent samples. Discrimination and selective-risk metrics use class-stratified resampling, which keeps the number of minority-class items fixed across draws and prevents an interval from being dominated by fluctuation in π_0 ; the routing analysis resamples routing items unstratified and recomputes the ordering within each draw. All resampling uses 2000 draws and fixed seeds recorded in the analysis source and output caches.

Two properties matter for reading the intervals. First, all fitted objects — head set, standardisation, residualiser, and probe ensemble — and their score vectors are frozen before resampling. The fixed-budget ranking and cutoff are reapplied to each resampled multiset. The intervals are therefore *conditional on the fitted ensemble* and do not include fitting variability. Second, fitting variability is reported separately rather than pooled: individual initialisation AUROCs have standard deviations of 0.003–0.014, and the train-resample spread (which additionally varies the sampled training set and the selected heads) is reported alongside it. Combining the two into one $\pm$ would answer neither question.

C. Why the operating point is a budget

An earlier analysis chose each threshold on train to match the judge’s training precision. That does not equalise test precision after prevalence or score drift, so the resulting recalls lie at different test operating points. We instead rank without test labels and require every detector to select the same M_m items; this is both operationally defined and comparable across detectors.

D. Proportion intervals

Recalls at a fixed budget are proportions of a fixed denominator (the number of errors in the analysed population), and their marginal intervals use the Wilson score interval rather than the Wald interval, which is unreliable at the small error counts of the low-prevalence GSM8K cells (as few as 80 errors for Gemma 3/GSM8K). Differences between detectors always use the paired bootstrap of Section B-B, never a difference of two marginal intervals.

E. What is not corrected

The six model–dataset cells constitute a family of comparisons and the reported intervals are per-cell and nominal. We do not apply a family-wise correction, and we do not want the reader to infer one has been applied. Similarly, the hyperparameter ranges of Section E-C are minima and maxima over a grid evaluated on the test split; they are descriptive extrema, not confidence intervals, and they do not license selecting a configuration post hoc. The MATH difficulty strata of Section E-E carry no interval at all.

F. Why no calibration metric is reported

Every quantity in this paper is a functional of the ordering induced by the score (Section III-B), so any strictly increasing reparametrisation leaves all results unchanged. The probe’s outputs are consequently *scores*, not calibrated probabilities, and reporting expected calibration error or a Brier score would attach probability semantics the construction does not support: the MLP oversamples the minority class and the logistic regression uses balanced class weights, both of which shift the output scale without a separate train-only calibration stage. A deployment that needs a calibrated probability rather than a ranking would have to add that stage and validate it separately.

APPENDIX C

AUDIT OF THE $p(\text{YES})$ BASELINE

The comparison in this paper is only as sound as its baseline, so the construction of Equation (13) is audited rather than assumed. Three things could make the baseline unfairly weak: a bad verdict-token rule, a distribution the deployer could not actually observe, and saturation that a careless cutoff would resolve arbitrarily. We measure all three.

A. The verdict-token rule

V_{YES} and V_{NO} are built by matching the verdict-position vocabulary against surface forms of the two verdicts. On a subword vocabulary this is error-prone in an uncontrolled direction: on Gemma 3’s vocabulary, 765 distinct tokens match a NO prefix under a naive rule, including " not" and " November". We therefore compare prefix matching against exact matching of the verdict tokens. The two agree to within 1.1×10^{-3} AUROC in the worst of the six cells (Llama 3.1/MATH, 0.6956 prefix versus 0.6946 exact) and to within 10^{-5} in the best, so the choice of rule does not carry any conclusion in the paper. The rule is pinned by a regression test and the per-model matched-token counts are recorded in the artifact manifest.

The recomputed $p(\text{YES})$ is additionally cross-checked against the value stored at generation time for every record; across all runs reported here the two agree exactly (maximum absolute difference 0). That check is not decorative — it caught a stale verdict-distribution dump that survived every configuration- and version-based cache check, because those fingerprint the registry entry and the metric formula but not the dump’s contents.

B. What a real API returns

A commercial API exposes only the top- K log-probabilities, typically $K = 20$, not the full next-token distribution, so a baseline computed from the full distribution is stronger than one a black-box deployer could build. We recompute Equation (13) under top-20 and top-40 caps, renormalising over the verdict tokens that survive the cap and assigning the neutral value 0.5 where neither appears.

The baseline is essentially unaffected: the largest AUROC difference between the capped and uncapped forms across the

six cells is 9.2×10^{-5} (Llama 3.1/GSM8K), and the fallback to 0.5 fires on at most 7 of 4883 records (Gemma 3/MATH) and on zero records in four of the six cells. The verdict tokens are, in practice, the top of the distribution at the verdict position. We report the uncapped form in the main text, which is the marginally stronger baseline and therefore the conservative choice for our claim.

C. Saturation and exact tie sizes

$p(\text{YES})$ is concentrated near the endpoints. Counting records with $p(\text{YES}) \leq 0.02$ or $p(\text{YES}) \geq 0.98$: 1270 of 1319 (Gemma 3/GSM8K), 1260 of 1319 (Qwen2.5/GSM8K), 977 of 1307 (Llama 3.1/GSM8K), 4654 of 4883 (Gemma 3/MATH), 4408 of 4949 (Qwen2.5/MATH), and 3039 of 4466 (Llama 3.1/MATH). Thus 68%–96% lie near an endpoint, but this saturation band is not an exact tie group.

The exact score audit gives 144, 137, and 688 unique values on the three GSM8K cells, and 418, 668, and 1686 on MATH (Gemma 3, Qwen2.5, Llama 3.1). The largest exact tie groups have 64, 58, 17, 173, 162, and 74 items respectively. At the 5% cutoff, the corresponding tie groups contain only 2, 2, 1, 1, 1, and 1 items; over all reported review budgets the largest cutoff tie contains 56 items. Exact tie-averaging therefore remains the reproducible convention, but its effect is localized rather than the thousand-item ambiguity previously implied by the saturation counts.

D. Why the sampled verdict is not the baseline

The binary verdict $v \in \{\text{YES}, \text{NO}\}$ is a thresholded read of Equation (13), and its ROC area is algebraically balanced accuracy, $\frac{1}{2}(\text{TPR} + \text{TNR})$. This is a valid AUROC for a two-level ranking, but it discards all within-verdict ordering. Using the operational $p(\text{YES})$ values reported in Table I, the gap between the two forms of the same signal is large: 0.703 versus 0.848 (Gemma 3/GSM8K), 0.582 versus 0.873 (Qwen2.5/GSM8K), 0.545 versus 0.697 (Llama 3.1/GSM8K), and 0.602, 0.614, 0.522 versus 0.822, 0.848, 0.696 on the three MATH runs. Comparing a probe against the verdict rather than against $p(\text{YES})$ would therefore credit the probe with up to 0.29 AUROC that is attributable to reading a distribution instead of a token. We therefore retain the verdict only as this diagnostic and use $p(\text{YES})$ for every operational comparison.

APPENDIX D

ANALYSED POPULATIONS AND EXCLUSIONS

Table VI gives the full exclusion accounting behind Section V-B. Three points about it are worth stating explicitly.

The exclusion changes the target population: A budget-truncated solution stops before writing its final answer, so it parses to no answer and would be scored wrong. Retaining these records would inflate the observed error prevalence — from 0.312 to 0.371 in the Llama 3.1/MATH train split, for instance — and add a potentially easy-to-detect population of texts that stop mid-sentence. Excluding them lowers prevalence in every affected cell (the π_0 versus π_0^+ columns), but

TABLE VI

ANALYSED POPULATION AND EXCLUSIONS, PER MODEL–DATASET CELL AND SPLIT. “TRUNC” IS GENERATIONS CUT OFF BY THE TOKEN BUDGET BEFORE WRITING A FINAL ANSWER; “NO GOLD” IS REFERENCE ANSWERS THAT DID NOT PARSE. π_0 IS THE ERROR PREVALENCE OF THE ANALYSED POPULATION AND π_0^+ WHAT IT WOULD BE IF TRUNCATIONS WERE RETAINED AND SCORED WRONG. EXCLUDING THEM LOWERS OBSERVED PREVALENCE IN EVERY AFFECTED CELL.

Run	split	gen.	trunc	no gold	analysed	π_0	π_0^+
Gemma 3 12B / GSM8K	train	7473	7	0	7466	0.057	0.057
	test	1319	0	0	1319	0.061	0.061
Qwen2.5 7B / GSM8K	train	7473	0	0	7473	0.050	0.050
	test	1319	0	0	1319	0.081	0.081
Llama 3.1 8B / GSM8K	train	7473	30	0	7443	0.107	0.110
	test	1319	12	0	1307	0.178	0.185
Gemma 3 12B / MATH	train	7500	197	2	7301	0.177	0.198
	test	5000	117	0	4883	0.184	0.203
Qwen2.5 7B / MATH	train	7500	83	2	7415	0.234	0.242
	test	5000	51	0	4949	0.270	0.277
Llama 3.1 8B / MATH	train	7500	645	2	6853	0.312	0.371
	test	5000	534	0	4466	0.494	0.548
total		63876	1676	6	62194		

the direction of its effect on the probe–judge difference is not known.

Llama’s MATH exclusions are non-termination, not long solutions: Llama 3.1 8B loses 8.6% of its MATH train split and 10.7% of its test split, by far the largest exclusion in the table, and this shifts its analysed prevalence from 0.312 (train) to 0.494 (test). A budget-calibration run finds that 37% of its truncated problems still exceed 12,288 tokens — three times the budget actually used — so these are generations that do not terminate rather than ordinary long solutions. Excluding them defines performance conditional on termination; a system-level evaluation could instead count non-termination as failure. The train/test prevalence shift nonetheless means Llama’s MATH probe is fitted at a materially different error rate than it is evaluated at, which is a caveat on that cell specifically.

Labels are recomputed, not trusted: Each record carries an advisory `labelable` field written at generation time, and the authoritative value is recomputed at analysis time. Files written before this was corrected recorded the field as “a gold answer exists”, ignoring truncation, and therefore disagreed with the analysis on exactly the truncated records. The loader audits the two and prints any divergence rather than allowing a hand inspection of the raw records to imply a different population than the figures were built from.

A. Answer equivalence

The relation $\equiv$ of Equation (1) is per-dataset and is recomputed from each raw record at analysis time.

GSM8K uses exact numeric equivalence after parsing, so 64, 64.0 and 64.00 are one answer.

MATH normalises $\LaTeX$ and then compares as strings, additionally accepting decimal equality when *both* sides are plain numeric literals. Thus $\backslash\mathrm{dfrac}\{1\}\{2\}$, $\backslash\mathrm{frac}12$ and $\backslash\mathrm{frac}\{1\}\{2\}$ are one answer, while $\backslash\mathrm{frac}\{1\}\{2\}$ and 0.5 are *not*. This strict rule may mislabel equivalent forms and is not accompanied by a symbolic-equivalence sensitivity.

The solve prompt requests the final answer inside $\backslash\mathrm{boxed}\{\}$, and reference answers are read from the last $\backslash\mathrm{boxed}\{\}$ of the reference solution; a reference that boxes nothing is unlabelable and excluded, whereas a *model* response that boxes nothing counts as wrong, because the reference answer is still known.

APPENDIX E ROBUSTNESS AND BOUNDS

These analyses support the main claims rather than carrying them. Each is labelled with what it can and cannot establish.

A. Full transfer matrix

Table VII gives every fit–target direction for all three detectors with paired intervals, of which Table III shows the primary probe, and Figure 5 plots it. Two patterns carry the argument of Section VII. The controlled probe’s loss exceeds the raw probe’s on the same items in every GSM8K→MATH direction, a pattern consistent with Lemma 1. And the shallow detector loses least of the three in all six directions, with three of its six loss intervals including zero.

B. Full risk–coverage profiles

Figure 6 gives the finite-sample selective-prediction view, $\mathcal{S}(c_m) = \varrho_m$, over the full coverage range. It is the same measurement as Figure 2 by Proposition 1 and is included because it is the conventional presentation and because it shows the whole curve rather than the deployable region. Note that reading a summary such as AURC off this figure integrates uniformly over coverage — the same objection Proposition 2 raises against AUROC.

C. Evidence consistent with a nonlinear read-out advantage

The three arbitrary defaults — $n_h = 32$ heads, $C = 0.5$, one hidden layer of 128 units — were varied one axis at a time ($n_h \in [8, 128]$, $C \in [0.05, 10]$, hidden sizes from (32) to (128, 128)) across all six model–dataset cells, giving 84 cells. The MLP beats logistic regression in all 84. Granting each probe family its most favourable cell anywhere in the grid — which hands LR its best configuration and is therefore the comparison least flattering to our claim — still leaves the MLP ahead by +0.054 to +0.156 AUROC.

Two details bear on the design choices rather than on the headline. Shrinking the hidden layer from 128 to 32 units costs under 0.009 AUROC, so the observed gap is not strongly sensitive to hidden width in this grid. The default $n_h = 32$ *understates* the linear probe: LR climbs monotonically as more heads are added while the MLP is flat in n_h , so a larger n_h would narrow the gap on the linear side rather than widen the nonlinear one. The comparison is consistent with a nonlinear read-out advantage, but MLP and LR also differ in ensembling, minority balancing, optimisation, and regularisation, so it does not isolate nonlinearity causally. The monotone behaviour of LR in n_h is also why we make no sparsity claim in Section IV-C.

TABLE VII

FULL TRANSFER MATRIX UNDER THE JOINT PROTOCOL CHANGE. COLUMNS AS IN TABLE III; † MARKS A PAIRED 95% INTERVAL EXCLUDING ZERO. THE CONTROLLED PROBE LOSES MORE THAN THE RAW PROBE ON THE SAME ITEMS IN EVERY GSM8K→MATH DIRECTION, A PATTERN CONSISTENT WITH LEMMA 1 BUT NOT A CAUSAL TEST; THE SHALLOW DETECTOR LOSES LEAST OF THE THREE EVERYWHERE.

Model	Target	judge	detector	in	→	Δ vs. in-dom. [95% CI]	Δ vs. judge [95% CI]
Gemma 3 12B	GSM8K	0.848	raw MLP	0.879	0.808	-0.071 [-0.109, -0.034]†	-0.040 [-0.077, -0.002]†
			controlled	0.844	0.802	-0.041 [-0.097, +0.017]	-0.046 [-0.093, -0.002]†
			shallow	0.756	0.738	-0.018 [-0.048, +0.013]	-0.109 [-0.168, -0.054]†
	MATH	0.822	raw MLP	0.914	0.808	-0.106 [-0.122, -0.091]†	-0.014 [-0.027, -0.001]†
			controlled	0.902	0.567	-0.335 [-0.360, -0.310]†	-0.255 [-0.280, -0.230]†
			shallow	0.801	0.801	-0.000 [-0.005, +0.005]	-0.021 [-0.035, -0.007]†
Qwen2.5 7B	GSM8K	0.873	raw MLP	0.858	0.819	-0.040 [-0.080, -0.001]†	-0.054 [-0.092, -0.020]†
			controlled	0.830	0.774	-0.057 [-0.109, -0.005]†	-0.100 [-0.148, -0.052]†
			shallow	0.735	0.711	-0.025 [-0.054, +0.002]	-0.162 [-0.208, -0.120]†
	MATH	0.848	raw MLP	0.888	0.763	-0.125 [-0.139, -0.110]†	-0.085 [-0.099, -0.070]†
			controlled	0.873	0.624	-0.249 [-0.268, -0.230]†	-0.224 [-0.244, -0.203]†
			shallow	0.802	0.772	-0.030 [-0.038, -0.022]†	-0.076 [-0.090, -0.062]†
Llama 3.1 8B	GSM8K	0.697	raw MLP	0.813	0.757	-0.055 [-0.096, -0.016]†	+0.060 [+0.023, +0.099]†
			controlled	0.785	0.723	-0.062 [-0.107, -0.016]†	+0.027 [-0.013, +0.066]
			shallow	0.717	0.692	-0.025 [-0.045, -0.005]†	-0.005 [-0.053, +0.045]
	MATH	0.696	raw MLP	0.865	0.595	-0.270 [-0.289, -0.251]†	-0.100 [-0.117, -0.084]†
			controlled	0.853	0.448	-0.405 [-0.426, -0.385]†	-0.247 [-0.266, -0.228]†
			shallow	0.780	0.771	-0.009 [-0.015, -0.003]†	+0.075 [+0.057, +0.094]†

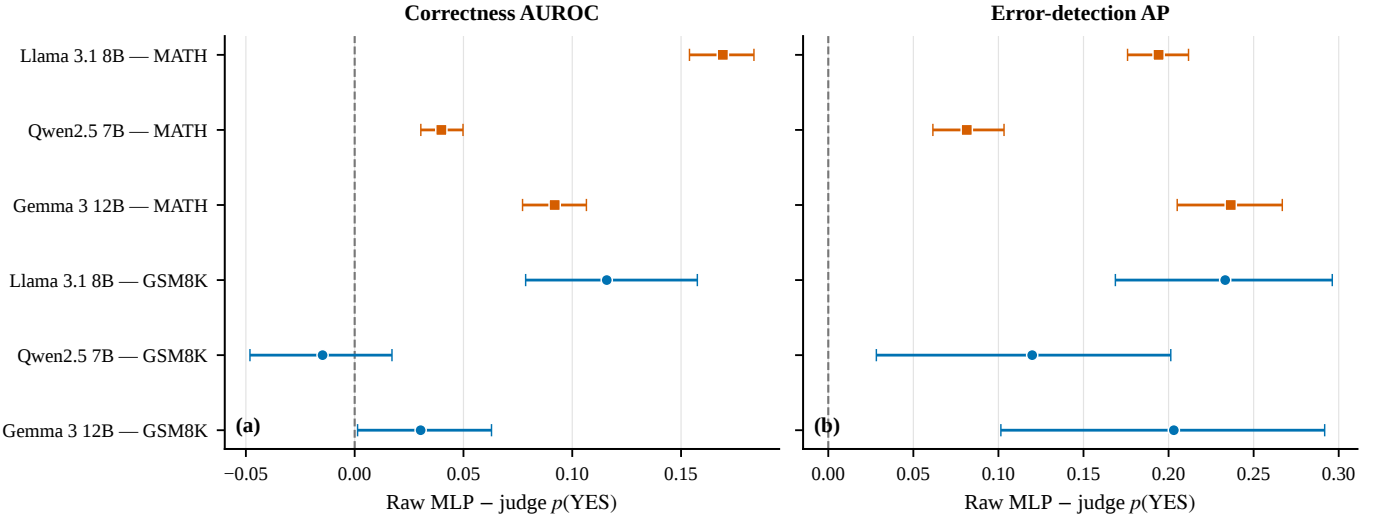


Fig. 4. Probe minus judge $p(\text{YES})$ on held-out correctness AUROC (left) and error-class average precision (right), with paired 95% class-stratified bootstrap intervals over test items. AP treats errors as the positive class and is therefore sensitive to the observed prevalence. Intervals are conditional on the fitted five-seed ensemble. These are the values tabulated in Table I.

D. Surface-form perturbation

GSM8K templates re-instantiated with fresh names and numbers give new benchmark instances, but their absence from pretraining cannot be established. Probes are fitted on the real GSM8K train split only and scored on the perturbed set. Primary-probe AUROC moves by -0.023 (Qwen2.5), $+0.029$ (Llama 3.1) and -0.011 (Gemma 3) against the full GSM8K test split, and solver accuracy moves by -4.6 , -0.7 and -1.8 points paired over templates.

Read this as a bound, not an exclusion.: Every one of those intervals covers zero and is roughly ± 0.1 AUROC wide,

because the perturbed side is 100 templates however many instances they generate. The perturbation rules out a *large* recognition effect and cannot rule out a small one; a few hundredths of AUROC would pass unseen. Tightening the bound requires more templates, not more instances per template. GSM-Symbolic’s templates also derive from GSM8K, so absence from pretraining is not established even for the perturbed set.

What is resolvable.: The probe-minus-judge margin on the perturbed set is paired row by row and is measured much more sharply than either detector’s absolute shift. For

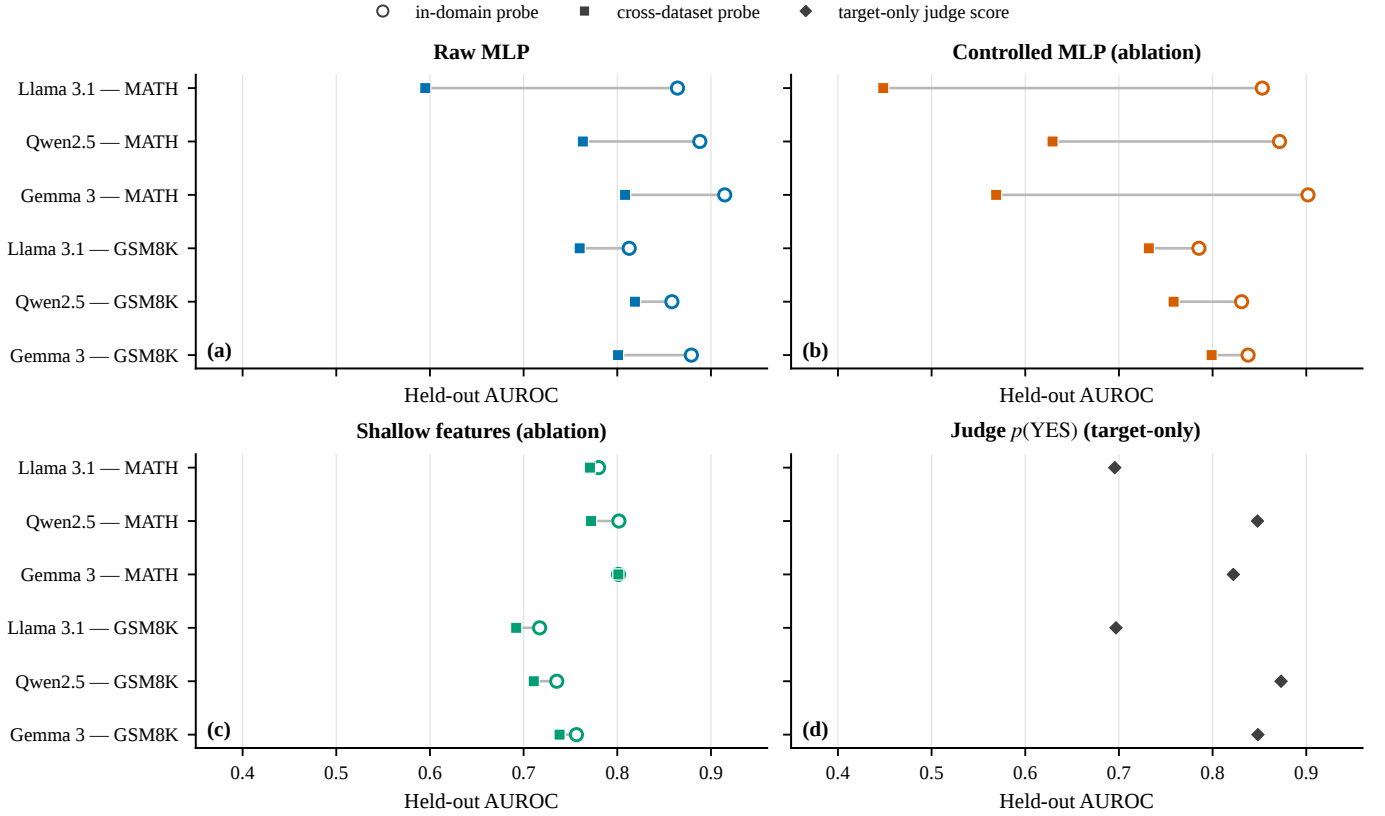


Fig. 5. Held-out target-split AUROC when each score is fitted in domain (open circle) or on the other mathematics dataset (filled square). Diamonds mark the target-domain judge $p(\text{YES})$, which needs no probe fitting. Paired intervals for every difference shown here are in Table VII.

Llama 3.1 it is $+0.147 [+0.085, +0.216]$, excluding zero: the probe’s advantage survives re-instantiation wherever it held on the original GSM8K.

An observation we decline to interpret.: The largest movement anywhere in the perturbation table belongs to Qwen2.5’s judge, whose $p(\text{YES})$ AUROC falls from 0.873 to 0.783 (-0.090). Its interval is $[-0.208, +0.016]$ and covers zero. The paired difference-in-differences against the probe, which cancels most of the resampling noise, is $+0.067 [-0.020, +0.141]$ and also covers zero; only against shallow text features does it exclude zero ($+0.119 [+0.032, +0.210]$), and that is partly because shallow features *gain* on the perturbed set while the judge loses. Llama 3.1 (0.697/0.696/0.694 across GSM8K, MATH and perturbed GSM8K) and Gemma 3 (0.848/0.822/0.865) are stable across all three. The supportable statement is that Qwen2.5’s judge behaves differently from every other detector under re-instantiation; benchmark familiarity is a hypothesis this experiment does not have the resolution to test.

E. Difficulty strata

Stratifying the MATH test splits by the dataset’s own difficulty level gives 15 strata (3 models $\times$ 5 levels, $n = 419$ to 1298) spanning error prevalences from 3.7% to 74.8%. The probe-minus-judge margin is positive in all 15, ranging from $+0.021$ (Qwen2.5, Level 4) to $+0.188$ (Llama 3.1, Level 1),

and it remains positive at Level 5 for every model ($+0.045$, $+0.154$, $+0.091$). Primary-probe AUROC itself is near flat across the range, from 0.810 to 0.936, despite the twentyfold change in prevalence.

This is descriptive: there are no level-wise intervals, the strata are not independent of the model’s own competence, and difficulty is the dataset’s annotation rather than a measured quantity. What it supports is a negative claim — the probe’s advantage is not confined to easy problems, and does not vanish where the model fails most — which is the relevant one for the contamination question of Section VIII.

APPENDIX F REPRODUCIBILITY AND DISCLOSURE

A. What is released

The public software and data artifact is available at <https://github.com/MakGulati/auditing-the-judge-artifact>. The fixed camera-ready release is linked from the start of this supplement. It contains the generation, judging, activation-extraction, probe-fitting, routing, transfer, and plotting source; pinned dependency manifests and tests; portable aligned held-out prediction arrays; cached statistics and rendered figures; and provenance metadata. It deliberately excludes the manuscript source, raw generation records, activation dumps, and model weights. The omitted experimental outputs are regenerable

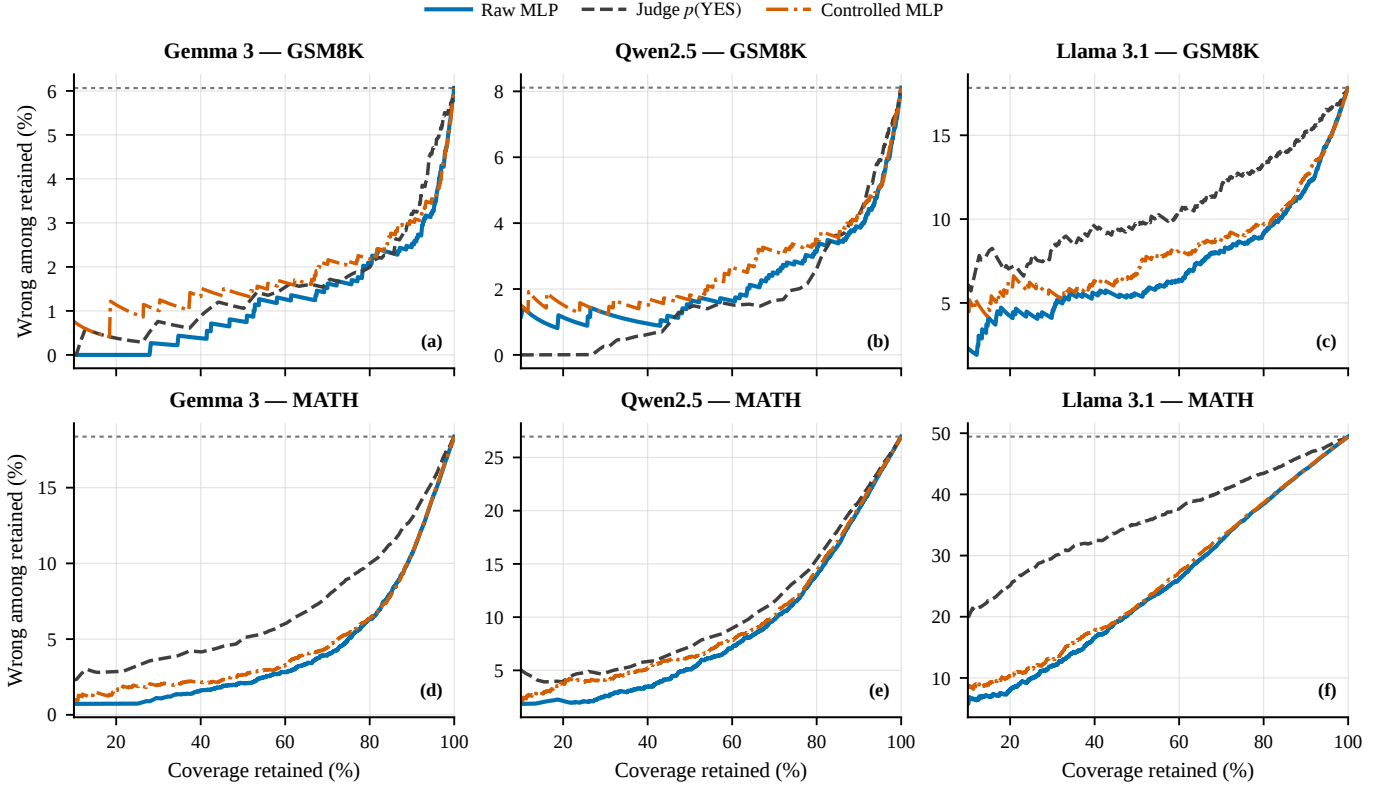


Fig. 6. Wrong-answer rate among retained answers as realised coverage $c_m = 1 - b_m$ increases from 10% to 100%; lower is better. The dashed line is the no-filtering prevalence, which every curve meets at $c = 1$ by construction. Ties use their exact expected risk under uniform random tie-breaking (Section B-A); exact cutoff-tie sizes are audited in Section C-C.

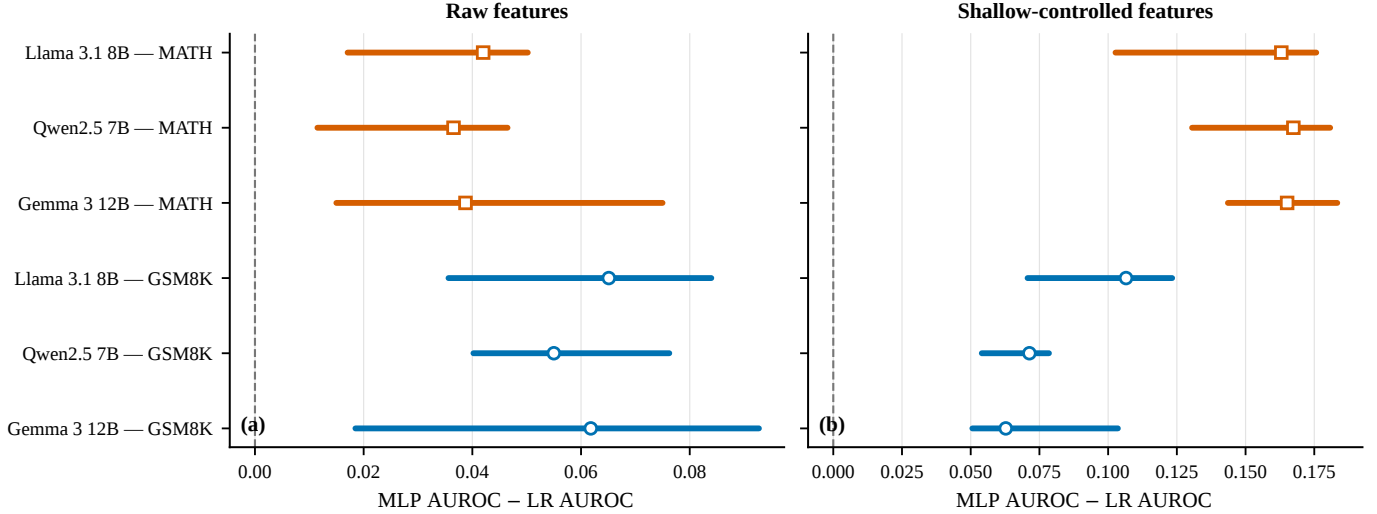


Fig. 7. Range of held-out MLP-minus-LR AUROC as the number of retained heads, regularisation, and hidden width vary; open circles mark the frozen default. These are descriptive test-set extrema, *not* confidence intervals, and they do not license selecting a configuration post hoc or attributing the gap to nonlinearity alone.

by the supplied scripts from the named upstream datasets and checkpoints, but are not redistributed because of size and upstream terms. The artifact therefore supports CPU reproduction of the reported numerical analyses and figures, as well as end-to-end regeneration when model access and

suitable GPU resources are available.

B. Provenance binding

Three checks bind an analysis to the generation run it claims to describe, and each was added after a failure it now prevents.

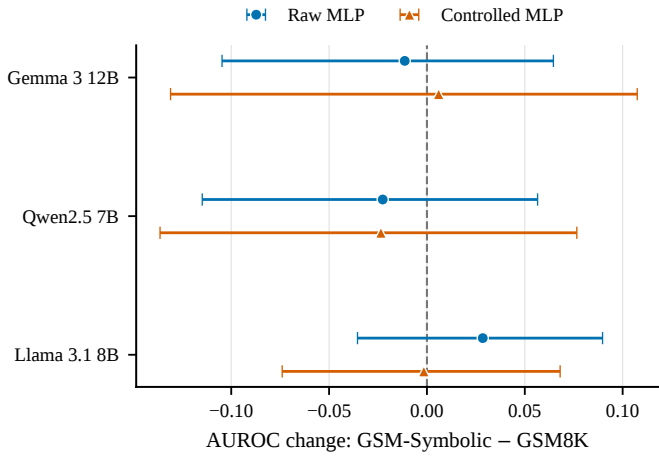


Fig. 8. AUROC difference between GSM-Symbolic and the full GSM8K test split. Whiskers independently resample template clusters on the perturbed side and rows on the full-test side; this is not a paired comparison. Intervals generally include zero, so the figure bounds only a large population difference and does not establish perturbation invariance.

Prompt reproduction. Activation extraction re-templates the judge prompt from scratch, so a different checkpoint, tokenizer, or assistant prefill would read activations at a different token position than the one that emitted the recorded verdict, with no visible symptom downstream. An extractor refuses to run when it cannot reproduce the prompt.

Row-level fingerprints. Every dump row carries a content fingerprint of the record it was extracted from, so a dump cannot be resumed or analysed against a raw file it did not come from even when the indices line up.

Geometry. Each dump records the model id and attention geometry, and the probe refuses a head dimension that disagrees with it. A wrong head dimension previously reshaped silently, fragmenting real attention heads into halves.

Dumps that predate these checks are refused by the loader rather than plotted, and this — not a selection decision — is why two earlier checkpoints are absent from Table I (Section V-A).

C. Two independent pipelines agree

The results in this paper were computed by a figure package that fixes the primary method in advance. A separate, earlier analysis pipeline computes overlapping quantities by independent code paths, and where the two overlap they agree: the 5%-budget recalls for Gemma 3/GSM8K are 0.4875 / 0.2625 / +0.225 in both; the routing gain over random at $m = 0.10$ for the controlled MLP is +2.05 points in both, with bootstrap intervals $[+1.20, +2.96]$ and $[+1.25, +3.00]$ differing only by resampling noise; and per-run cached AUROC agrees to 6×10^{-5} (raw) and 2×10^{-3} (controlled). We report this because a cross-implementation check is evidence that the numbers are properties of the data rather than of one script.

D. Known sources of run-to-run variation

MLP fits vary by about 0.002 AUROC between otherwise identical runs, traceable to BLAS threading; this is below the reported initialisation spread of 0.003–0.014 but above zero, so numbers recomputed by a different script may differ in the third decimal. Greedy decoding is only about 47% byte-reproducible across runs at all, which bounds what any single-run absolute accuracy claim can mean and is why the $k = 1$ and $k = 5$ populations are reported separately (Section VIII). The rebuilt probe scores use a newer scikit-learn than the earlier caches; the per-run metadata records the full analysis stack for both. The reported generation runs used a single NVIDIA RTX 5000 Ada Generation GPU with 32 GiB memory; the pinned vLLM environment was verified with driver 555.

E. Probe optimisation settings

The head covariance uses ridge $\lambda = 10^{-2}$. Logistic regression uses ℓ_2 regularisation with $C = 0.5$ and balanced class weights. The MLP uses an ℓ_2 penalty of 10^{-4} , learning rate 10^{-3} , at most 300 iterations, and a 15% internal early-stopping split; its minority class is oversampled with replacement. The five ensemble initialisations use seeds 0–4.